

Roll No.

TBT-701

**B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

IPR, BIOSAFETY & BIOETHICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Differentiate between biosafety and biosecurity with examples.

(CO5, CO2, CO3)

(b) Highlight the role and functions of World Trade Organization.

(CO5, CO2, CO3)

(c) How do the biological resource centers help in the maintenance of sustainability ? Give a detailed explanation.

(CO5, CO2, CO3)

P. T. O.

2. (a) Explain the risks associated with the infectious agents. Also give a brief explanation on the risk levels created for the effective handling of the pathogenic strains. (CO3, CO5)
- (b) "An effective and environmentally sensitive approach to pest management that relies on a combination of common-sense practices." Comment on this statement. (CO3, CO5)
- (c) Describe the components of biorisk management along with the risk mitigation strategies. (CO3, CO5)
3. (a) Discuss the registration procedure for trademark and explain the infringement strategies followed for trademark infringement. (CO1, CO5)
- (b) Give the detailed account of registration of any plant variety. Also define the characteristics of a plant variety which holds the property of being protected. (CO1, CO5)
- (c) Explain the methodology adopted for transportation of hazardous materials. (CO1, CO5)
4. (a) Explain the concept of bioethics with relevance to the existing medical techniques. (CO4, CO2)
- (b) Discuss the salient features of TRIPS. Also highlight its significance. (CO4, CO2)
- (c) Describe any one case study in which the issues pertaining to the infringement of copyright exists. (CO4, CO2)

(3)

5. (a) "The right prohibits the extraction or re-utilisation of any database in which there has been a substantial investment in obtaining, verifying or presenting the data contents". Comment on the statement. (CO1, CO2)
- (b) Describe the sections which deal with the patentable and non-patentable subject matter. (CO1, CO2)
- (c) Differentiate between Tariff and Quotas with examples. (CO1, CO2)

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TBT-702(B)

B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

RATIONAL DRUG DISCOVERY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. Describe various steps of Drug discovery using computational biology.

(CO1)

(b) Explain any *one* tool and its application — PyRx OR Biovia Discovery Studio.

(CO1)

(c) Describe various methods used for drug target identification and validation.

(CO1)

2. (a) Differentiate between Quantum Mechanics and Molecular Mechanics.

(CO2)

(b) Explain MM3, OPLS, AMBER and CHARMM. Which equation is used for calculation of total force fields ?

(CO2)

P. T. O.

(2)

- (c) Why do we apply energy minimization ? Discuss Steepest Descent algorithm for Energy minimization. (CO2)
3. (a) Discuss the algorithm of Molecular dynamics simulation. What do you mean by Periodic boundary conditions ? (CO3)
- (b) Explain the following file formats : (CO3)
- (i) .trr
 - (ii) em.log
 - (iii) .xvg
 - (iv) minim.mdp
 - (v) nvt.tpr
- (c) What do you mean by Principal component analysis ? Describe major steps in PCA. (CO3)
4. (a) Which rule is used to assess ADMET parameters in a potential molecule ? Explain different phases of Clinical trial of drug molecules. (CO4)
- (b) Explain any *two* protein sequences as well as structure databases. (CO4)
- (c) Describe parameters of docking. What is significance of RMSD in docking ? (CO4)
5. (a) Differentiate between 2D and 3D OSAR. What are *four* main parameters of QSAR analysis ? (CO5)
- (b) Describe Artificial Neural Network and applications of ANN in drug designing. (CO5)
- (c) What is a pharmacophore ? Discuss pharmacophore modelling in detail. (CO5)

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TBT-702(C)

B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

ENZYME TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Elaborate on any *five* characteristic properties of enzymes. (CO1)
(b) Explain the different scopes of Enzyme Technology. (CO1)
(c) Define chromatography. Describe in detail the technique of Affinity Chromatography. (CO1)
2. (a) Derive mathematically the Michaelis–Menten rate equation. (CO2)
(b) Explain in detail the following : (CO2)
 - (i) Enzyme Kinetics
 - (ii) Enzyme substrate complex

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(2)

- (c) Explain with the help of a diagram in detail the various features of a substrate velocity plot. (CO2)
3. (a) Describe the mechanism of action of the Chymotrypsin enzyme. (CO3)
- (b) What is activation energy ? Explain how enzymes can be used as catalysts to lower the activation energy. (CO3)
- (c) Elaborate on the competitive and non-competitive inhibition. (CO3)
4. (a) Explain the role of enzymes in the production of wine and beer. (CO4)
- (b) Elaborate on the use of enzymes in the health and pharmaceutical industry. (CO4)
- (c) Discuss the role of enzymes in the bakery and dairy industry. (CO4)
5. (a) Elaborate on enzyme engineering and discuss its applications and advantages. (CO5)
- (b) What is enzyme immobilization ? Explain the gel entrapment method for enzyme immobilization in detail. (CO5)
- (c) Elaborate on the methods of Isolation, screening, and maintenance of industrially important microorganisms. (CO5)

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TBT-703(A)

B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

PHARMACEUTICAL BIOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the sources of drug ? Explain. (CO1)
(b) Describe the different routes of drug administration. (CO1)
(c) Describe the importance of parenteral route of drug administration. (CO1)
2. (a) What is difference between pharmacokinetics and pharmacodynamics ? (CO2)
(b) Describe the factor affecting route of drug administration. (CO2)
(c) Discuss drugs targeting with examples. (CO2)
3. (a) What is the meaning of bioavailability ? Explain. (CO3)

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(2)

- (b) Describe the production and analysis of biopharmaceuticals. (CO3)
- (c) Describe the good manufacturing facilities and source of biopharmaceuticals. (CO3)
- 4. (a) Describe the role of biopharmaceutical in treatment of various health disorders. (CO4)
- (b) Discuss in detail about different therapeutic proteins. (CO4)
- (c) Describe TNF and its role in biopharmaceuticals. (CO4)
- 5. (a) What is drug delivery system ? Explain in detail. (CO5)
- (b) Explain the concept of plant bioreactor expressing protein. (CO5)
- (c) Explain the liposome mediated drug delivery. (CO5)

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B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

PYTHON

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is Python, and why is it considered a popular programming language ? (CO1)
- (b) Can you explain some key features of Python that make it a versatile language for various applications ? (CO3)
- (c) What are data types in Python ? Why data types are important in any programming ? (CO4)
2. (a) Explain the purpose of an "if" statement in Python and how it is used to make decisions in code. Write a python code to accept person's age and the print if the person is eligible to vote or not ? (CO1)
(Voter : age $\geq$ 18 years)

P. T. O.

(2)

- (b) Differentiate between “for” and “while” loops in Python. When is each type of loop most commonly used ? Write a python code to print the first 10 even numbers. (CO4)
- (c) Define a function in Python and explain why they are important in programming. Write a python function to return the sum of digits of a given input integer. (CO5)
3. (a) What is a Iteration statement ? Write a Python program to reverse an Integer number. (CO1)
- (b) What is a list ? Explain some basic operations you can perform on a Python list, such as adding, removing, and accessing elements. (CO3)
- (c) What is a tuple in Python ? How does tuple differ from a list ? Write python code to illustrate the tuple. (CO5)
4. (a) What is a function ? State the benefits of writing functions. Explain function with default arguments (giving a suitable example). (CO1)
- (b) Explain the basic syntax for defining a function in Python. Write a Python function that calculates the factorial of a given number. (CO3)
- (c) What is a module ? Create a Python module that contains a function to calculate the square of a number. Import this module and use the function in your code. (CO4)
5. (a) What is Biopython ? How does it relate to the field of bioinformatics ? List some of the main modules of Biopython. (CO1)
- (b) Explain the concept of sequence objects in Biopython and why they are useful in bioinformatics. (CO3)
- (c) What is a sequence ? Utilize Biopython to compare two sequence objects and determine their similarity. (CO4)

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TBT-705(A)

B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

BIOTERRORISM AND NATIONAL SECURITY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Explain bioterrorism, its agents, limitations, and regulations for controlling its use. (CO1)
- (b) Describe the CBRN weapons with *one* example each. (CO1)
- (c) What is terrorism ? Define its objectives, goal, and history. (CO1)
2. (a) Explain the *two* cases of chemical weapons where mass destruction occurred. (CO2)
- (b) What is bacteria ? Describe the interaction of bacteria with immune system by taking one example. (CO2)

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(2)

- (c) How biological war agents impact the physiology of a human ? Give *one* example. (CO2)
3. (a) How do you define any microbial outbreak as natural or targeted ? Give example. (CO3)
- (b) What are the key reasons of using any microbe as biowar agents ? Describe the best biowar agent dispersal means. (CO3)
- (c) Explain why anthrax and botulism are preferred biological war agents. (CO3)
4. (a) Elaborate the vaccination supplies concept by taking any example. (CO4)
- (b) Describe the biowar agent detection systems. Mention their drawbacks and suggestion the modifications. (CO4)
- (c) Explain the role of international regulations in controlling the Bioterrorism. (CO4)
5. (a) Discuss the ethical and national policies on controlling bioterrorism. (CO5)
- (b) How you can create awareness among public about bioterrorism and national security ? (CO5)
- (c) What is microbial forensics ? How it is helpful in managing bioterrorism ? (CO5)

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TBT-705(B)

B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

FOOD AND NUTRITION SCIENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. Discuss *ten* historical breakthrough events of food science and technology.

(CO1)

(b) Discuss different intrinsic and extrinsic factors affecting interaction of microorganism with food.

(CO1)

(c) Discuss the term food spoilage. Discuss spoilage of milk and milk products in detail.

(CO1)

2. (a) Discuss various steps involved in production of beer.

(CO2)

(b) Discuss various microorganisms involved in spoilage of meat and egg.

(CO2)

P. T. O.

(2)

- (c) Write short notes on the following : (CO2)
- (i) Pro- and Pre-biotics
 - (ii) Starter Cultures
3. (a) Illustrate different emerging preservation technologies for milk and dairy products. (CO3)
- (b) Discuss different methods involved in microbiological examination of foods. (CO3)
- (c) Differentiate between terms flavor enhancer and flavoring agent. Discuss different sources of flavoring agents in detail. (CO3)
4. (a) Discuss the term food irradiation and working mechanism of both ionizing and non-ionizing radiation. (CO4)
- (b) Discuss classification of irradiation on the basis of dosage utilized and their purpose. (CO4)
- (c) Write short note on preservation of food by low temperature. (CO4)
5. (a) Discuss GMO and various ethical issues related to them. (CO5)
- (b) Discuss, symptoms, food responsible and their precaution for the following food borne infections : (CO5)
- (i) Salmonellosis
 - (ii) Clostridium perfringens gastroenteritis
 - (iii) Botulism poisoning
- (c) Discuss different future applications of food biotechnology. (CO5)